

Control of a 9-Phase PMSM with Stacked Polyphase Bridge Converter including DC Source Impedance

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Index Terms—Control of drive, Multiphase drive, Multiphase converter, Modular converter, Integrated motor drives.

Abstract—This paper presents a control concept for a drive system consisting of a 9-phase PMSM connected to a stacked polyphase bridge converter (SPBC) under consideration of the DC source impedance (R_b, L_b). When the converter is connected to the DC source with a cable, the cable impedance represents a source impedance that could cause instabilities affecting the total DC link voltage and the module voltage balance. In this paper, a control concept is presented for stabilising the module voltage balance via an appropriate definition of the module reference voltage. The total DC link voltage is not controlled but stability is ensured by sufficiently large DC link capacitors. The complete drive controller concept is validated in a simulation of a 1.5kW drive system.

I. INTRODUCTION

A promising solution to fulfil the need for more efficient, smaller, and less expensive electrical drives is the concept of integrated motor drives (IMDs) [1]. IMDs combine an electric motor and its driving inverter in one mechanical unit. A converter topology that is particularly suited to IMDs is the stacked polyphase bridge converter (SPBC, [2]) due to its modularity and due to its lower dv/dt on the motor windings compared to conventional two-level topologies [3]. Furthermore, the SPBC can be used in combination with multiphase motors. Multiphase motor windings such as for example 9-phase motor windings enable a higher winding factor than conventional 3-phase windings. The higher winding factor results in a higher efficiency/power density of the drive system [4]. In addition, DC cables are used to connect IMDs to a DC voltage source that is located with some distance to the IMDs. Possible applications

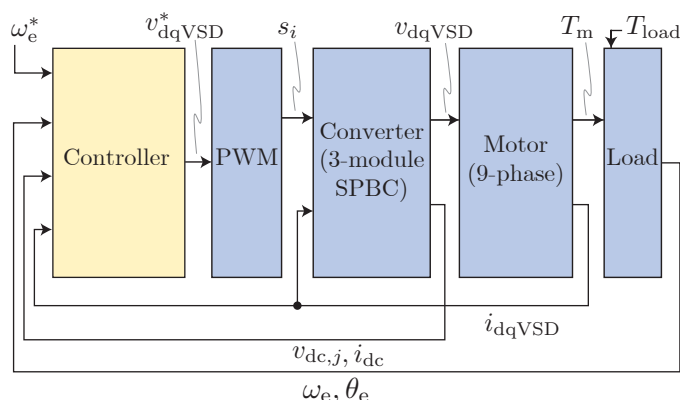


Fig. 1: Considered drive system configuration. The system blocks in blue represent the plant of the closed loop control system.

for such cables are electric cars (for connecting the inverter(s) to the battery) or factories with a single DC supply connected to multiple distributed IMDs.

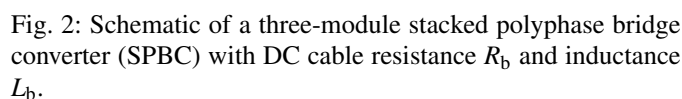
The control of a drive system including multiphase motors, SPBCs, and DC cables is a challenging task because it requires a non-standard transformation matrix for the motor current control, DC link balancing, and total DC link voltage stabilisation.

In the literature several techniques for the current/torque control of multiphase PMSMs are presented, as reviewed in [5]. More recent contributions to such control techniques are given in [6–10]. However, the presented techniques assume inverter topologies that do not require DC link balancing. Control concepts considering both current/torque control and DC link balancing control in a modular series connected inverter are given in [11–16]. However, [11–15] implicitly assume phase-aligned 3-phase winding modules without any magnetic coupling, whereas multiphase windings such as 9-phase windings consist of non-phase-aligned 3-phase winding modules that are magnetically coupled. This magnetic coupling requires a different current control concept. A control concept combining such current control with voltage

- 1) How should the reference voltage for the DC link modules be defined? In [16], the total DC link voltage V_{dc} is considered constant. In that case the obvious choice is $V_{dc}/3$, as the number of 3-phase modules is 3. However, as shown in Fig. 2, the converter model with source impedance (R_b, L_b) has two DC link voltages, V_{dc} at the supply and v_{Σ} at the converter. Hence, $V_{dc}/3$ or $v_{\Sigma}/3$ are potential candidates for the reference voltage. [14] shows that the choice of the reference voltage affects the system stability when all module DC link voltages are controlled. However, in [16] only two out of three module DC link voltages are controlled. Hence the results from [14] cannot directly be applied.
- 2) For a given definition of the module reference voltage, under what conditions is the system stable?
- 3) Given that the control concept from [16] involves multiple control loops in a nested structure, under what assumptions can each controller be tuned separately using standard control design techniques?

In the following, first a model of the controlled drive system, i.e. of the *plant* in terms of control theory is given in section II. This includes the blue system blocks in Fig. 1 showing the complete drive system configuration. The structure of the controller (yellow block in Fig. 1) is presented in section III. There, the module DC link reference voltage is defined as $v_{\Sigma}/3$. The closed loop stability of the voltage balancing control is analysed in section IV, also leading to an explanation for why $v_{\Sigma}/3$ is the appropriate reference voltage. Taking the resulting stability condition into account, a controller tuning method is given in section V. Finally, the controller concept is validated in a drive system simulation for an exemplary 1.5kW PMSM in section VI.

This section presents the models of the drive system plant, corresponding to the blue system blocks in Fig.1,



A. Converter model

$$V_{\text{dc}} = R_{\text{b}} i_{\text{dc}} + L_{\text{b}} \frac{di_{\text{dc}}}{dt} + v_{\Sigma}. \quad (1)$$
$$v_{\Sigma} = \sum_{j=1}^3 v_{\text{dc},j}. \quad (2)$$
$$C \frac{dv_{dc,j}}{dt} = i_{c,j} = i_{dc} - i_{dc,j}. \quad (3)$$
$$i_{\text{dc},j} = \frac{P_j}{v_{\text{dc},j}} \quad (4)$$

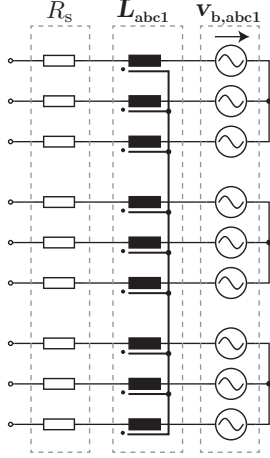


Fig. 3: Model of a 9-phase PMSM in the abc1-phase space, including the phase resistance R_s , the inductance matrix L_{abc1} , and the back electromotive force vector $v_{b,abc1}$.

where P_j is the power flowing from the j -th DC link capacitor to the j -th 3-phase converter module. By assuming a high converter efficiency, the power P_j approximately corresponds to the power flowing into the motor. This approximation is made for the stability analysis of the closed loop voltage control in section IV.

B. PWM model

As PWM, standard third harmonic injection pulse width modulation (THIPWM, section 5.3 of [17]) is used. For the controller structure and for the controller tuning the PWM is approximated as ideal, whereas for the simulation presented in section VI the PWM/switching is taken into account.

C. Motor model and used reference frames

The considered symmetrical 9-phase motor model is the same as the one considered in [16] except for an adaptation from induction machines to PMSMs, which does not affect the control concept, but only the motor voltage equations. The motor voltage equations can be expressed in different variable spaces/reference frames as is explained in the following.

The control concept in [16] involves six different reference frames/variable spaces, which are called abc, abc1, VSD, MS, dqVSD, and dqMS. The abc-space is a phase space where the phases are sorted by increasing phase angle. The abc1-space is a phase space where the phases are sorted by groups of 3-phase winding systems. The VSD-space is a variable space in which the voltage equations of the considered 9-phase machine are decoupled. The MS-space is a variable space in which the power consumed/generated by each 3-phase module

can be expressed. The dqVSD-space and the dqMS-space are rotating reference frames in which sinusoidal VSD-/MS-space voltages/currents are constant.

The control concept presented in section III of this paper only uses variables in the variable spaces abc, abc1, dqVSD, and dqMS. The transformation matrices required to transform voltages/currents between these variable spaces are given in the appendix in section VIII. In the equations of this paper, the respective variable space used for expressing motor voltages/currents is indicated by subscripts using the following notation for a general 9-phase voltage/current vector x in the abc-, dqVSD-, abc1-, or dqMS-space:

$$\begin{bmatrix} x_{abc} \\ x_{dqVSD} \\ x_{abc1} \\ x_{dqMS} \end{bmatrix}^T = \begin{bmatrix} x_a & x_{ds} & x_{a1} & x_{d1} \\ x_b & x_{qs} & x_{b1} & x_{q1} \\ x_c & x_{dx1} & x_{c1} & x_{z1} \\ x_d & x_{qy1} & x_{a2} & x_{d2} \\ x_e & x_{dx2} & x_{b2} & x_{q2} \\ x_f & x_{qy2} & x_{c2} & x_{z2} \\ x_g & x_{z1} & x_{a3} & x_{d3} \\ x_h & x_{z2} & x_{b3} & x_{q3} \\ x_i & x_{z3} & x_{c3} & x_{z3} \end{bmatrix} \quad (5)$$

The dqVSD-space is used for the current controllers presented in section III. The current controllers are based on the motor voltage equations adapted to PMSMs in the dqVSD-space, resulting in (6)-(9).

$$\begin{bmatrix} v_{ds} \\ v_{qs} \end{bmatrix} = \begin{bmatrix} R_s + L_s \frac{d}{dt} & -\omega_e L_s \\ \omega_e L_s & R_s + L_s \frac{d}{dt} \end{bmatrix} \begin{bmatrix} i_{ds} \\ i_{qs} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_e \psi_{PM} \end{bmatrix} \quad (6)$$

$$\begin{bmatrix} v_{dx1} \\ v_{qy1} \end{bmatrix} = \begin{bmatrix} R_s + L_{ls} \frac{d}{dt} & \omega_e L_{ls} \\ -\omega_e L_{ls} & R_s + L_{ls} \frac{d}{dt} \end{bmatrix} \begin{bmatrix} i_{dx1} \\ i_{qy1} \end{bmatrix} \quad (7)$$

$$\begin{bmatrix} v_{dx2} \\ v_{qy2} \end{bmatrix} = \begin{bmatrix} R_s + L_{ls} \frac{d}{dt} & -\omega_e L_{ls} \\ \omega_e L_{ls} & R_s + L_{ls} \frac{d}{dt} \end{bmatrix} \begin{bmatrix} i_{dx2} \\ i_{qy2} \end{bmatrix} \quad (8)$$

$$v_{z,j} = \left(R_s + L_{ls} \frac{d}{dt} \right) i_{z,j}, \quad j = 1, 2, 3 \quad (9)$$

There, R_s is the motor winding resistance, L_s and L_{ls} are the main and leakage inductance, ψ_{PM} is the permanent magnet flux linkage, and ω_e is the electrical angular rotational frequency of the rotor, also called rotor speed in the following. It must be noted that the voltage equations in the dqVSD-space (6)-(9) assume an equal winding resistance R_s for all phase windings and a symmetrical winding distribution. In order to test the control with disturbance given by unequal resistances, a motor model in the abc1-phase space, shown in Fig. 3, is used for deriving the simulation results presented in section VI.

The dqMS-space is used to express the power consumed

by the j -th 3-phase motor winding. If a high converter efficiency is assumed, this power is approximately equal to the power P_j (Fig. 2) flowing out of the j -th DC link module capacitance. Assuming motor current control with maximum torque per ampere (MTPA) and a non-salient rotor, the motor current vector $\mathbf{i}_{dqMS}$ only has non-zero q-components (x_{q1}, x_{q2} , and x_{q3} in (5) where $\mathbf{x}$ is given by $\mathbf{i}$), which results in the following expression for the power P_j

$$P_j = \frac{3}{2} R_s i_{q,j}^2 + \frac{3}{2} \psi_{PM} i_{q,j} \omega_e. \quad (10)$$

D. Load model

The load model of the drive system in Fig. 1 is given by the following standard equation of motion (Tab. 3.1 of [18]),

$$J \frac{d\omega_e}{dt} = p(T_m - T_{load}) \quad (11)$$

where J is the moment of inertia connected to the rotor, T_m is the mechanical motor torque, T_{load} is the load torque, and p is the pole pair number of the motor. As indicated in Fig. 1, the outputs of the load model are the rotor speed $\omega_e = d\theta_e/dt$ and the rotor position θ_e . In the model θ_e results from a double integration of (11). In the real system θ_e is measured using an encoder.

III. CONTROLLER STRUCTURE

In this section the structure of the proposed controller for the plant of the drive system, described in the previous section, is presented. Therefore, the top level control and its included controllers for the DC link balancing, currents, and speed are explained. The focus lies on the DC link balancing controller.

A. Top level control

Fig. 4 shows the top level structure of the drive controller, which corresponds to the top level structure proposed in [16], with the additional voltage signal v_Σ (Fig. 2) that is fed into the DC link balancing controller. The main concept is to control the motor torque and the module DC link voltages with current components of different dqVSD-subspaces which means different groups of vector components of the motor current vector $\mathbf{i}_{dqVSD}$. Among all $\mathbf{i}_{dqVSD}$ -components given in (5), the current components i_{qs} & i_{ds} (forming a subspace called $idqs$) are used for the torque control, whereas the current components i_{dx1} & i_{qy1} (forming a subspace called $dqxy1$) and the current components i_{dx2} & i_{qy2} (forming a subspace called $dqxy2$) are used for the DC link balancing control. The currents of the subspaces dqs,

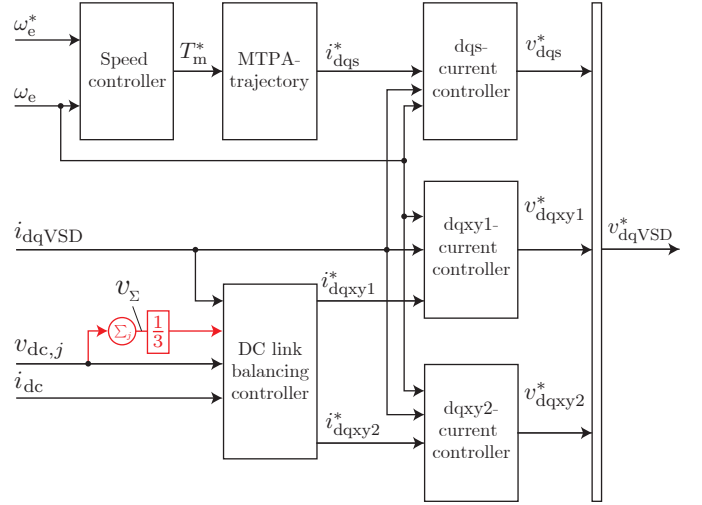


Fig. 4: Top level structure of the drive controller based on [16], corresponding to the yellow block in Fig. 1. The voltage signal $v_\Sigma/3$ is added as input to the DC link balancing controller.

dqxy1, and dqxy2 are controlled separately within the dqs-, dqxy1-, and dqxy2-current controller blocks shown in Fig. 4.

B. DC link balancing controller

Fig. 5 shows the proposed DC link balancing controller. The balancing strategy from [16] is kept, which is to add a positive or negative balancing current $i_{q,j,bal}$ to $i_{q,j}$ when the module voltage $v_{dc,j}$ is larger or lower than its reference value $v_{dc,j}^*$. Hence (with (10)), the power drawn from the j -th module is increased or decreased, which in turn decreases or increases the module voltage $v_{dc,j}$ if the machine is operating in motor mode ($i_{dc} > 0$) and vice versa. In addition, to avoid affecting the torque, the sum of the reference balancing currents $i_{q,j,bal}^*$ for all modules ($j = 1, 2, 3$) is set equal to zero by the controller in Fig. 5 by only controlling two out of the three module voltages and by setting the third reference balancing current $i_{q3,bal}^*$ to $(-i_{q1,bal}^* - i_{q2,bal}^*)$. The red parts in Fig. 5 show the changes with regard to [16] explained in the following:

- 1) In the proposed controller implementation, the current references in the dqxy(1,2)-subspaces are directly calculated in the dqMS- and in the dqVSD-space, without any intermediate step involving variables in the VSD-space, which requires less transformations. Furthermore, the presented controller calculates the current references based on Δk_j instead of $k_j = 1 + \Delta k_j$, which is possible because the added "1" is cancelled out in the dqMS/dqVSD-

transformation block in Fig. 5. This change also reduces the computational effort.

- 2) In the proposed controller implementation, the reference voltage $v_{dc,j}^*$ is chosen such that the effect of the source impedance (R_b, L_b) is taken into account: The reference value is set to $v_\Sigma/3$. In section IV, this choice for $v_{dc,j}^*$ is shown to be correct in the sense that it leads to an equivalent controller action on all DC link module voltages despite the fact that only two out of three voltages are controlled. It is noted that with this choice, the controller in Fig. 5 is a *pure balancing* controller, i.e. it does not act on the total DC link voltage v_Σ .

A total DC link voltage controller could theoretically be added with additional balancing currents $i_{q,j, \text{bal}}$ that are defined equal for all modules $j = 1, 2, 3$ and that are set in the dqS-subspace. However, such a total DC link voltage controller, that would provide active damping of the LC resonant circuit at the inverter input, is practically hardly feasible/useful because of the following reasons:

- 1) A total DC link voltage controller would have to control the total power flow out of the DC link (i.e. the total motor power) and hence the motor currents with a bandwidth of the order of magnitude of the resonance frequency $f_{\text{res}} = 1/\sqrt{L_b C/3}$. For reasonable values of L_b and C , e.g. using the specifications given in Tab. I, the resonance frequency results in $f_{\text{res}} \approx 100\text{kHz}$. Given that the current controller bandwidth is limited by the PWM frequency f_{PWM} , a total DC link voltage control would require $f_{\text{PWM}} \gg 100\text{kHz}$ and would hence result in high switching losses.
- 2) Controlling the total power flow of the motor leads to additional torque ripple.

For these reasons, a total DC link voltage controller is not considered further.

C. Dqs-current controller

Fig. 6 shows the dqs-current controller from Fig. 4, controlling the current vector components i_{ds} & i_{qs} . The structure is that of the well known cross-coupling decoupling controller for electrical drives, here adapted to PMSMs, which leads to the ψ_{PM} -block in Fig. 6.

D. Dqxy(1,2)-current controllers

The structure of the two dqxy(1,2)-current controllers in Fig. 4 is shown in Fig. 7. This structure incorporates two modifications compared to the complex vector PI

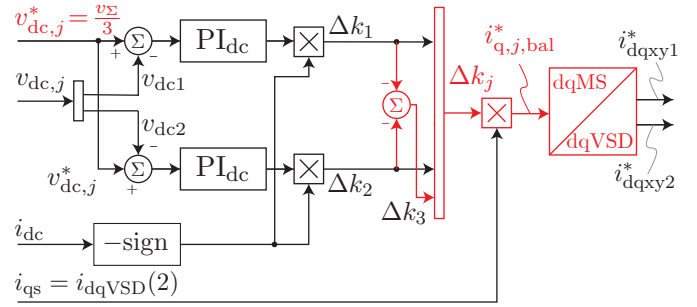


Fig. 5: Proposed structure of the DC link balancing controller. Changes compared to [16] are marked in red: 1) The reference balancing currents $i_{q,j, \text{bal}}^*$ are directly calculated from Δk_j without any intermediate step involving $k_j = 1 + \Delta k_j$, leading to less computational effort. 2) The appropriate module DC link reference voltage with present DC link cable impedance (R_s, L_s) is added.

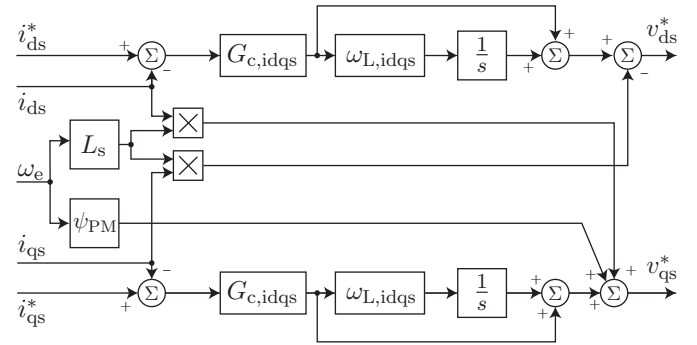


Fig. 6: Controller for the currents i_{ds} and i_{qs} of the torque producing dqs-subspace. The controller structure corresponds to the well known PI controller with cross-coupling decoupling.

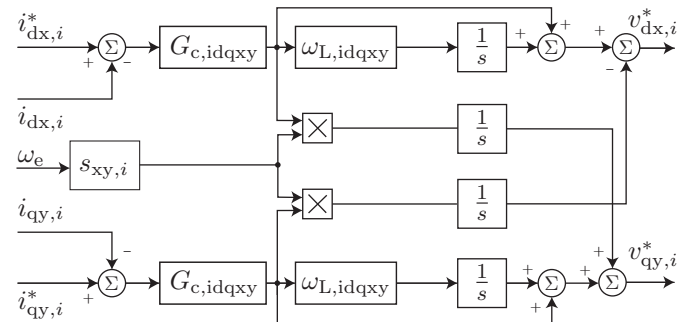


Fig. 7: Controller for the currents $i_{dx,i=(1,2)}$ & $i_{qy,i=(1,2)}$ of the subspaces dqxy(1,2). There, $s_{xy,i} = \{-1, 1\}$ for $i = \{1, 2\}$. The controller is based on a complex vector PI controller [16], split up into its real and imaginary part and generalised with regard to the corner frequency $\omega_{L, \text{idqxy}}$ that is kept general.

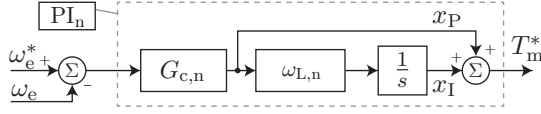


Fig. 8: PI controller for the motor speed.

controller given in [16]: 1) The controller from [16] is based on a block diagram using complex numbers. In contrast, the block diagram in Fig. 7 only uses real numbers by splitting up the complex vector PI controller into its real and imaginary part. This form of the controller is required for the implementation in a real system and for the simulation presented in section VI. 2) The controller shown in Fig. 7 has been generalised with respect to the corner frequency $\omega_{L, \text{idqxy}}$ which is kept general in order to separate the controller structure from the controller tuning that is considered later in section V.

The reason for using the controller from Fig. 7 in the dqxy(1,2)-space instead of the cross-coupling decoupling controller from Fig. 6 is that the dqxy(1,2)-currents depend on the relatively small leakage inductance L_{ls} and that the controller from Fig. 7 is less susceptible to estimation errors of L_{ls} (section 3.6 of [19]).

E. Speed controller

Fig. 8 shows the speed controller which is a simple PI controller that sets the torque reference T_m^* based on the speed error. Fig. 8 also serves as a model for all used PI_x -controllers with individual high frequency gain $G_{c,x}$ and corner frequency $\omega_{L,x}$, where x corresponds to idqs, idqxy, dc, or n, for the dqs-current controller, the dqxy(1,2)-current controllers, the DC link balancing controller, or the speed controller, respectively.

IV. STABILITY ANALYSIS

Among the controllers described in the previous section, the main focus of the current controllers (dqs, dqxy1, dqxy2) and of the speed controller is on reference tracking and disturbance rejection whereas the focus of the DC link balancing controller is on stabilisation because the converter/DC link plant model given by (1)-(4) can be open-loop unstable in motor mode ($P > 0$) [14]. The series DC link system has two groups of eigenvalues (called *instability modes* in [14]), one caused by the resonance of L_b and the total DC link capacitance $C/3$, and one caused by the instability of the module voltage balance [14]. The way to stabilise these groups of eigenvalues is determined based on the method from [14]. The required steps for the stability analysis are:

- 1) Linearisation of the DC link state space model (SSM) given by (1)-(4) with the states $\Delta v_{dc,j}$ and

Δi_{dc} and with the inputs ΔP_j . The result of the linearisation is given in (9)-(12) of [14]. The linearisation process involves additional parameters for the equilibrium/steady state operating points for which the equations are linearised. In the following, these parameters are v_0 , P_0 , i_{qs0} , and ω_{e0} , defining the steady state values of $v_{dc,j}$, P_j , $i_{q,j}$, and ω_e , respectively.

- 2) By adding the sum of the differential equation (3) for $j = 1, 2, 3$ to the system of equations from step 1, the system from step 1 is transformed into two subsystems, as shown in Fig. 9: 1) A *tot-system* for the total DC link voltage with the states Δv_Σ and Δi_{dc} and with the inputs ΔP_j . 2) A *mod-system* for the module DC link voltages with the states $\Delta v_{dc,j}$ and the inputs Δi_{dc} and ΔP_j . The SSM equations are given in (12) for the *tot-system* and in (13) for the *mod-system*.

$$\dot{\Delta i}_{dc} = -\frac{R_b}{L_b} \Delta i_{dc} - \frac{1}{L_b} \Delta v_\Sigma \quad (12a)$$

$$\dot{\Delta v}_\Sigma = \frac{3}{C} \Delta i_{dc} + \frac{P_0}{C v_0^2} \Delta v_\Sigma - \frac{1}{C v_0} \sum_{j=1}^3 \Delta P_j \quad (12b)$$

$$\dot{\Delta v}_{dc,j} = \frac{P_0}{C v_0^2} \Delta v_{dc,j} + \frac{1}{C} \Delta i_{dc} - \frac{1}{C v_0} \Delta P_j \quad (13)$$

- 3) Linearisation of the input power equation (10) resulting in (14).

$$\Delta P_j = \underbrace{\frac{3}{2} (2R_s i_{qs0} + \omega_{e0} \psi_{PM})}_{:=a} \Delta i_{q,j} \quad (14)$$

- 4) Formulation of the *current setting strategy* by expressing $\Delta i_{q,j}$ as a function of the voltage error ($v_{dc,j}^* - v_{dc,j}$) as defined by the DC link controller structure, here given in Fig. 5, resulting in (15).

$$\Delta i_{q,j|j=1,2} = -((\Delta v_{dc,j}^* - \Delta v_{dc,j}) G_{c,dc} + \Delta x_{I,j}) i_{qs0} \quad (15a)$$

$$\Delta i_{q3} = -\Delta i_{q1} - \Delta i_{q2} \quad (15b)$$

$$\dot{\Delta x}_{I,j|j=1,2} = G_{c,dc} \omega_{L,dc} (\Delta v_{dc,j}^* - \Delta v_{dc,j}) \quad (15c)$$

There, $x_{I,j}$ is the output/state of the integrator of the PI_{dc} -controller in Fig. 5. The condition $i_{dc} > 0$ is assumed as the system can only be open-loop unstable for motor mode [14]. By defining the reference voltage $v_{dc,j}^*$ as $\frac{v_\Sigma}{3}$ and by inserting (15a) into (15b), (15) results in (16).

$$\Delta i_{q,j|\forall j} = -\left(\left(\frac{\Delta v_\Sigma}{3} - \Delta v_{dc,j}\right) G_{c,dc} + \Delta x_{I,j}\right) i_{qs0} \quad (16a)$$

$$\dot{\Delta x}_{I,j|\forall j} = G_{c,dc} \omega_{L,dc} \left(\frac{\Delta v_\Sigma}{3} - \Delta v_{dc,j}\right) \quad (16b)$$

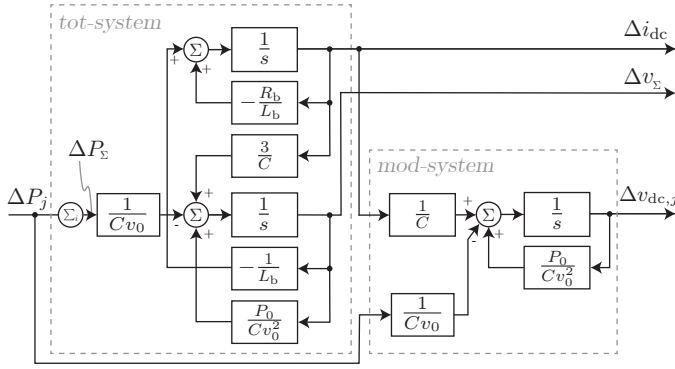


Fig. 9: Block diagram showing the *tot-system* state space model (12) and the *mod-system* state space model (13).

This results in the desired behaviour because the controller balances all modules equally since the form of (16) is equal for all modules $j = 1, 2, 3$. Therefore $v_\Sigma/3$ is the correct choice for $v_{dc,j}^*$.

- 5) Insertion of the current setting strategy, (16), and of ΔP_j , (14), into the SSMs of the *tot-system*, (12) and of the *mod-system*, (13), and application of the Routh-Hurwitz stability criterion for each resulting closed loop SSM, called *cl-tot-system* and *cl-mod-system*. According to the Routh-Hurwitz stability criterion, asymptotic stability is given when all coefficients of the characteristic polynomial $c(s) = \det(s\mathbf{I} - \mathbf{A})$ are positive, where $\mathbf{A}$ is the state matrix of the considered closed loop SSM and $\mathbf{I}$ is the unity matrix. The results of the stability analysis are: The *cl-tot-system* is the same as the (open-loop) *tot-system* because the sum of the module powers

$$\sum_{j=1}^3 \Delta P_j = a \underbrace{\sum_{j=1}^3 \Delta i_{q,j}}_0 \quad (17)$$

is zero. This result validates that the defined DC link controller does not act on the total DC link voltage Δv_Σ . Hence, the stability criterion for the total DC link stability corresponds to the open loop stability criterion given in [14],

$$C > \frac{P_0 L_b}{v_0^2 R_b} \quad (18)$$

which means that the total DC link stability is ensured by using a large enough DC link capacitor. For the *cl-mod-system*, the resulting SSM is different from the open-loop system and is given in matrix form in (19).

$$\underbrace{\begin{bmatrix} \Delta v_{dc,j} \\ \Delta x_{1,j} \end{bmatrix}}_{\Delta \mathbf{x}} = \underbrace{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & 0 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \Delta v_{dc,j} \\ \Delta x_{1,j} \end{bmatrix}}_{\Delta \mathbf{x}} + \underbrace{\begin{bmatrix} b_{11} & b_{12} \\ 0 & b_{22} \end{bmatrix}}_{\mathbf{B}} \underbrace{\begin{bmatrix} \Delta i_{dc} \\ \Delta v_\Sigma \end{bmatrix}}_{\Delta \mathbf{u}} \quad (19a)$$

$$a_{11} = \frac{1}{Cv_0} \left(\frac{P_0}{v_0} - ai_{qs0} G_{c,dc} \right) \quad (19b)$$

$$a_{12} = \frac{ai_{qs0}}{Cv_0}, a_{21} = -G_{c,dc} \omega_{L,dc} \quad (19c)$$

$$b_{11} = \frac{1}{C}, b_{12} = \frac{ai_{qs0} G_{c,dc}}{3Cv_0}, b_{22} = \frac{G_{c,dc} \omega_{L,dc}}{3} \quad (19d)$$

The characteristic polynomial of the state matrix $\mathbf{A}$ from (19) is then given by

$$\begin{aligned} c(s) &= \det(s\mathbf{I} - \mathbf{A}) \\ &= s^2 + s \left(\frac{ai_{qs0} G_{c,dc}}{Cv_0} - \frac{P_0}{Cv_0^2} \right) + \frac{ai_{qs0} G_{c,dc} \omega_{L,dc}}{Cv_0}. \end{aligned} \quad (20)$$

Applying the Routh-Hurwitz stability criterion results in the following stability condition

$$G_{c,dc} > \frac{1}{v_0} \left(1 - \frac{3R_s i_{qs0}}{2a} \right) \approx \frac{1}{v_0} \quad (21)$$

for the HF gain of the PI_{dc} -controller $G_{c,dc}$. There, the integrator has no effect on the stability, as $G_{c,dc} \omega_{L,dc} > 0$ is always ensured. It must be noted that the result in (21) corresponds to the submodule DC link stability criterion calculated in [14]. However, the results given in [14] are not directly applied here because the controller structure is different: The presented controller uses two PI-controllers for the control of three modules, whereas [14] uses one P-controller per module.

V. CONTROLLER TUNING

In this section, a procedure for tuning/setting the high frequency gains $G_{c,x}$ and the corner frequencies $\omega_{L,x}$ of the involved PI-controllers PI_x is presented. As mentioned in section III-E, x corresponds to id_{qs} , id_{qxy} , dc , or n , for the dqs -current controller, the $dqxy(1,2)$ -current controllers, the DC link balancing controller, or the speed controller, respectively. The procedure is based on splitting the complete control system into four control loops that, via the assumption/condition of sufficiently large frequency intervals between the bandwidths of the control loops, can be independently tuned. The control loops are shown in Fig. 10 and these result in a nested structure as the loops in Fig. 10c & Fig. 10d are part of the control loops in Fig. 10a & Fig. 10b, respectively. The tuning procedure is shown in Fig. 11. The first step

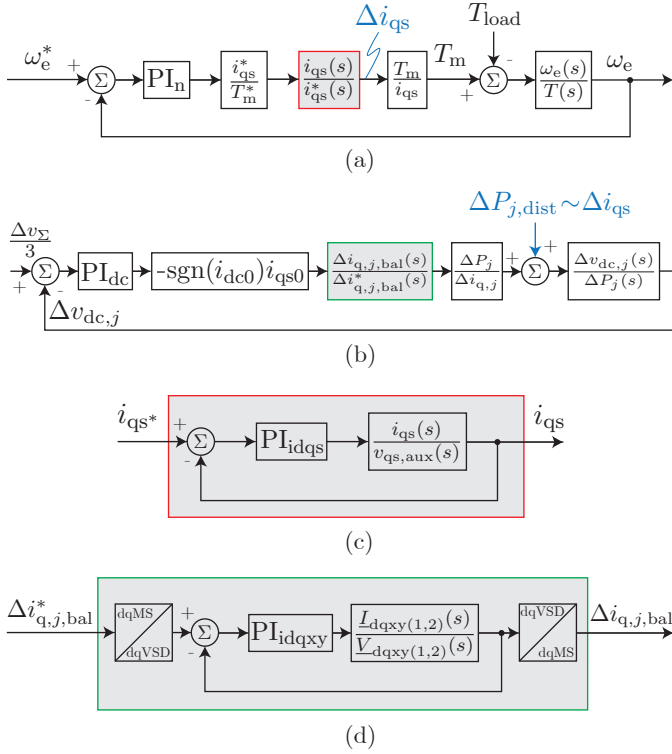


Fig. 10: Simplified linear model of the drive system used for the controller tuning, involving four feedback control loops: (a) Speed control loop. (b) DC link balancing control loop for $v_{dc,j}$ (identical control loop for each module). (c) Idqs-current control loop. (d) Idqxy(1,2)-current control loops.

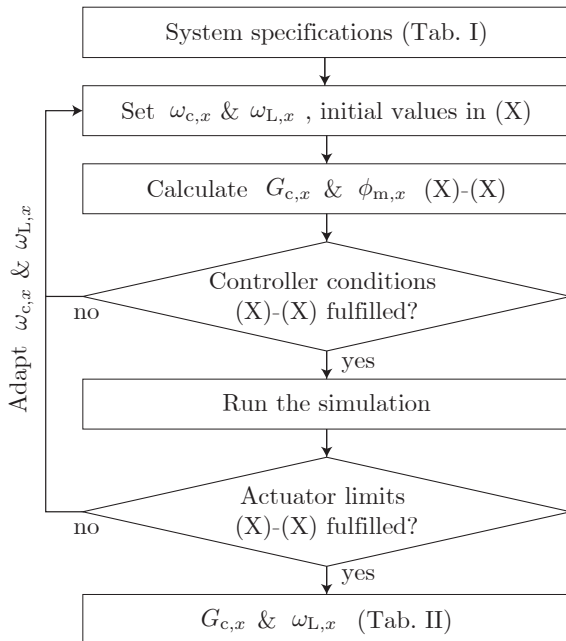


Fig. 11: Tuning procedure for the PI-controllers of the different control loops shown in Fig. 10.

of this procedure is to define the system specifications that are summarised in Tab. I. Then, the control loop bandwidths $\omega_{c,x}$ and the corner frequencies $\omega_{L,x}$ are set to the following initial values:

$$\omega_{c,idqxy,0} = 2\pi \frac{f_{PWM}}{10}, \quad \omega_{L,idqxy,0} = \frac{R_s}{L_{ls}} \quad (22a)$$

$$\omega_{c,dc,0} = \frac{\omega_{c,idqxy,0}}{10}, \quad \omega_{L,dc,0} = \frac{\omega_{c,dc,0}}{10} \quad (22b)$$

$$\omega_{c,idqs,0} = \omega_{c,dc,0}, \quad \omega_{L,idqs,0} = \frac{\omega_{c,idqs,0}}{10} \quad (22c)$$

$$\omega_{c,n,0} = \frac{\omega_{c,idqs,0}}{10}, \quad \omega_{L,n,0} = \frac{\omega_{c,n,0}}{10}. \quad (22d)$$

These initial values are chosen based on the controller conditions that are explained later in this section. In the next step, the high frequency gains $G_{c,x}$ are calculated using (23),

$$G_{c,idqxy} = L_{ls} \omega_{c,idqxy} \quad (23a)$$

$$G_{c,idqs} = L_s \omega_{c,idqs} \quad (23b)$$

$$G_{c,n} = \frac{\omega_{c,n} J}{p} \quad (23c)$$

$$G_{c,dc} = \frac{2\omega_{c,dc} CV_{dc}}{9 \left(2R_s i_{qs0}^2 + \psi_{PM} i_{qs0} \omega_{e0} \right)} \quad (23d)$$

and the phase margins $\phi_{m,idqs}$ and $\phi_{m,n}$ are calculated using (24),

$$\phi_{m,idqs} = 180^\circ + \angle T_{idqs}(s)|_{s=j\omega_{c,idqs}} \quad (24a)$$

$$\phi_{m,n} = 180^\circ + \angle T_n(s)|_{s=j\omega_{c,n}}. \quad (24b)$$

There, $T_{idqs}(s)$ and $T_n(s)$ are the open loop transfer functions of the idqs-current control loop and of the speed control loop which are given by

$$T_{idqs}(s) = G_{c,idqs} \left(1 + \frac{\omega_{L,idqs}}{s} \right) \frac{1/L_s}{s + \frac{R_s}{L_s}} \quad (25a)$$

$$T_n(s) = G_{c,n} \left(1 + \frac{\omega_{L,n}}{s} \right) \frac{p}{sJ}. \quad (25b)$$

In the next step, the controller parameters $G_{c,x}$ and $\omega_{L,x}$ are evaluated with regard to the following criteria:

- **Criterion 1:** The control loops should be decoupled. This criterion is fulfilled when the bandwidths of the inner control loops in Fig. 10c & Fig. 10d, $\omega_{c,idqs}$ and $\omega_{c,idqxy}$, are significantly larger than the bandwidths of the respective outer control loops in Fig. 10a & Fig. 10b, $\omega_{c,n}$ and $\omega_{c,dc}$, which is achieved by fulfilling the conditions

$$\omega_{c,n} \leq \frac{\omega_{c,idqs}}{10}, \quad \omega_{c,dc} \leq \frac{\omega_{c,idqxy}}{10}. \quad (26)$$

In addition, the bandwidth of the DC link balancing control loop (Fig. 10b) $\omega_{c,dc}$ should be larger than the bandwidth of the idqs-current control loop (Fig. 10a) $\omega_{c,idqs}$. This ensures a sufficient rejection of the disturbance $\Delta P_{j,dist}$, that results from a resistance imbalance and from the current Δi_{qs} that is set by the idqs-current control loop.

- **Criterion 2:** All control loops must be stable. The stability criterion of the DC link balancing control loop is already given in (21). The dqz-current control loop and the speed control loop are *single input single output* (SISO) systems that are open loop stable. Therefore, their closed loop stability can be ensured by the standard phase margin criterion for PI-control loops given by (section 9.4.1 of [20])

$$\phi_{m,idqs} > 0, \quad \phi_{m,n} > 0. \quad (27)$$

The dqxy(1,2)-current control loops, split up into their real and imaginary parts, are *multiple input multiple output* (MIMO) systems. Therefore, the same stability analysis, that is applied to the DC link balancing controller in section IV, is applied to the idqxy(1,2)-current controllers. The resulting condition for stability is that the PI-parameters $G_{c,idqxy}$ and $\omega_{L,idqxy}$ must be positive, which is always fulfilled by definition of a PI-controller.

- **Criterion 3:** The PWM can be considered ideal. This criterion requires a sufficient margin between the bandwidths of the current control loops $\omega_{c,idqs}$ & $\omega_{c,idqxy}$ and the PWM frequency f_{PWM} . This margin is commonly chosen as one decade, resulting in the conditions

$$\omega_{c,idqs} \leq 2\pi \frac{f_{PWM}}{10}, \quad \omega_{c,idqxy} \leq 2\pi \frac{f_{PWM}}{10}. \quad (28)$$

When the above mentioned conditions are fulfilled, the simulation is executed and the simulation result is checked for the controller output limits. The output limit of the current control loops is determined by the THIPWM-unit whose output voltage magnitude $v_{dq,j} = \sqrt{v_{d,j}^2 + v_{q,j}^2}$ for each three-phase module ($j = 1, 2, 3$) is limited by

$$v_{dq,j} \leq \frac{v_{dc,j}}{\sqrt{3}}. \quad (29)$$

The speed control loop and the DC link balancer control loop are limited by a maximum allowed transient phase

Table I: Exemplary specifications of a drive system with a high-torque low-speed PMSM.

Parameter	Symbol	Value
Rated power	$P_{m,r}$	1.5 kW
Rated torque	$T_{m,r}$	48 Nm
Rated speed	$\omega_{e,r}$	$2\pi \cdot 60 \text{ Hz}$
Rated current (rms)	I_r	2.4 A
Max. current (pk)	i_{max}	$2\sqrt{2}I_r$
Pole pair number	p	12
DqVSD-main inductance	L_s	64.7 mH
DqVSD-leakage inductance	L_{ls}	12.9 mH
Winding resistance	R_s	1.02 Ω
Magnet flux linkage	ψ_{PM}	262 mWb
DC link voltage	V_{dc}	800 V
DC link module capacitance	C	35 μF
DC cable inductance	L_b	7.5 μH
DC cable resistance	R_b	0.86 Ω
PWM frequency	f_{PWM}	20 kHz
Power for PI _{dc} design	P_0	$P_{m,r}/2$
Speed for PI _{dc} design	ω_{e0}	$\omega_{e,r}/2$

Table II: Controller parameters resulting from the tuning procedure in Fig. 11 applied to the specifications in Tab. I.

Parameter	Symbol	Value
HF gain of PI _{idqs}	$G_{c,idqs}$	45.4 V A ⁻¹
HF gain of PI _{idqxy}	$G_{c,idqxy}$	158.4 V A ⁻¹
HF gain of PI _{dc}	$G_{c,dc}$	85.2 mV ⁻¹
HF gain of PI _n	$G_{c,n}$	154 mNm s rad ⁻¹
Corner frequency of PI _{idqs}	$\omega_{L,idqs}$	49.1 rad s ⁻¹
Corner frequency of PI _{idqxy}	$\omega_{L,idqxy}$	79.1 rad s ⁻¹
Corner frequency of PI _{dc}	$\omega_{L,dc}$	614 rad s ⁻¹
Corner frequency of PI _n	$\omega_{L,n}$	0 rad s ⁻¹

current i_{max} in order to stay within the inverter/motor temperature limits. This results in the condition

$$i_a, i_b, \dots, i_i \leq i_{max}. \quad (30)$$

If the controller conditions (26)-(28) and the controller output limits (29)-(30) are fulfilled, a valid/working set of controller parameters is found. Otherwise, the controller bandwidths $\omega_{c,x}$ and corner frequencies $\omega_{L,x}$ must be adapted in an iterative procedure as shown in Fig. 11. It must be noted that a valid set of controller parameters can be further adapted/optimised with regard to system dynamics, e.g. based on the methods described in [21], which is however not within the scope of this paper.

VI. VALIDATION OF THE CONTROLLER

The control concept is validated in a simulation for the considered specifications summarised in Tab. I. The used simulation model corresponds to the model shown

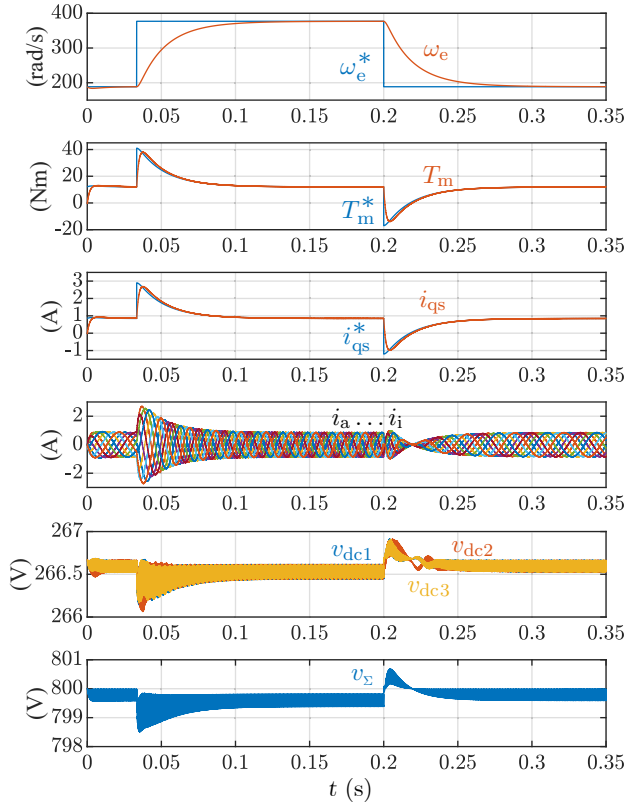


Fig. 12: Simulation results for a drive system with the proposed control concept and the specifications from Tab. I, showing (top to down) 1) the speed ω_e , 2) the torque T_m , 3) the current i_{qs} , 4) the motor currents in the abc-phase space, 5) the module DC link voltages $v_{dc,j}$, and 6) the total DC link voltage v_Σ .

in Fig. 1, where the converter and the motor are modelled by the electrical circuits shown in Fig. 2 and Fig. 3 and where the other parts (controller, PWM, and load) are modelled based on block diagrams. The block diagrams of the controller are given in Fig. 4-8. The block diagram of the PWM unit and of the load are not shown but the used models are explained in sections II-B & II-D. The controller parameters that are used in the simulation and that result from the tuning procedure in Fig. 11 are summarised in Tab. II. To increase the simulation speed, dead-time and measurement delays are not considered. Fig. 12 shows the simulation result for a speed step from 150rpm to 300rpm and back to 150rpm with a load torque of 12Nm. Similar to the approach in [16], a disturbance for the DC link voltage control is incorporated via a higher winding resistance of the second module, set to $2R_s$. The result in Fig. 12 shows good reference tracking of the speed in terms of ω_e and of the torque producing current i_{qs} , hence of the torque T_m as well. The DC link voltages $v_{dc,j}$ are well balanced.

The total DC link voltage v_Σ has peaks where the speed steps occur due to the cable impedance. However, these peaks are relatively small due to a sufficiently large DC link capacitance of 35 μ F (Tab. I).

VII. CONCLUSION

A control concept for drive systems consisting of a symmetrical 9-phase PMSM connected to an SPBC via a cable with the impedance (R_b , L_b) is presented. The concept is based on [16], adapted to PMSMs, and extended by introducing the cable impedance that represents a source impedance of the DC source. For taking this source impedance into account, the appropriate choice for the module DC link voltage reference is found to be $v_\Sigma/3$. This choice leads to a voltage controller only acting on the balancing of the DC link module voltages, independently of fluctuations of the total DC link voltage (v_Σ) during for example load steps. The total DC link voltage is kept stable via sufficiently large DC link capacitors. Controlling also the total DC link voltage is theoretically possible with an additional voltage controller in the torque producing current subspace. However, such total DC link voltage control is not considered practically useful due to its high bandwidth requirement and due to an additionally resulting torque ripple. The stability analysis of the module DC link voltages leads to a stability condition on the high frequency gain of the DC link balancing controller. Taking the stability condition for the gain into account, the controllers of the proposed concept are tuned in the frequency domain based on the assumption/condition of sufficiently large frequency intervals between the bandwidths of the involved control loops. Using the resulting controller parameters, the complete control concept is validated in a simulation for exemplary system specifications (Tab. I) based on a high-torque low-speed 1.5kW PMSM.

VIII. APPENDIX

In this section, the transformation matrices presented in [16] are summarised since these are required to transform any voltage/current vector $\mathbf{x}_\gamma$ in a given vector space, referred to as γ , into the voltage/current vector $\mathbf{x}_v$ in a different vector space, referred to as v . The general transformation is given by $\mathbf{x}_v = \mathbf{T}_{\gamma-v} \cdot \mathbf{x}_\gamma$. There, $\mathbf{T}_{\gamma-v}$ is the transformation matrix from vector space γ to vector space v . The vector spaces γ and v can be any of the vector spaces introduced in section II-C, i.e.

$\gamma, v = \{abc, \text{VSD}, \text{dqVSD}, \text{abc1}, \text{MS}, \text{dqMS}\}$.

The abc-to-VSD transformation matrix is given by

$$\mathbf{T}_{\text{abc-VSD}} = \frac{2}{9} \begin{bmatrix} \cos(\theta_s) \\ \sin(\theta_s) \\ \cos(2\theta_s) \\ \sin(2\theta_s) \\ \cos(4\theta_s) \\ \sin(4\theta_s) \\ 3/2\mathbf{I}_3 & 3/2\mathbf{I}_3 & 3/2\mathbf{I}_3 \end{bmatrix} \quad (31)$$

where $\mathbf{I}_3$ is the identity matrix with 3 rows/columns and θ_s is the vector of the winding axes angles of a symmetrical nine-phase motor given by

$$\theta_s = \frac{2\pi}{9} [0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8]. \quad (32)$$

The VSD-to-dqVSD transformation matrix is given by

$$\mathbf{T}_{\text{VSD-dqVSD}} = \begin{bmatrix} \mathbf{T}_R(-\theta_e) & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{T}_R(+\theta_e) & \mathbf{0} & \vdots \\ \vdots & \mathbf{0} & \mathbf{T}_R(-\theta_e) & \mathbf{0} \\ \mathbf{0} & \dots & \mathbf{0} & \mathbf{I}_3 \end{bmatrix} \quad (33)$$

where $\mathbf{T}_R(\theta_e)$ is the standard rotation matrix given by

$$\mathbf{T}_R(\theta_e) = \begin{bmatrix} \cos(\theta_e) & -\sin(\theta_e) \\ \sin(\theta_e) & \cos(\theta_e) \end{bmatrix}. \quad (34)$$

The abc-to-abc1 transformation matrix is given by

$$\mathbf{T}_{\text{abc-abc1}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (35)$$

which corresponds to a reordering of phases based on the grouping into three-phase winding systems.

The abc1-to-MS transformation matrix is given by

$$\mathbf{T}_{\text{abc1-MS}} = \begin{bmatrix} \mathbf{C}(\alpha=0) & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}(\alpha=\frac{2\pi}{9}) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{C}(\alpha=\frac{4\pi}{9}) \end{bmatrix} \quad (36)$$

where $\mathbf{C}(\alpha)$ is a generalised form of the Clarke transformation given by

$$\mathbf{C}(\alpha) = \frac{2}{3} \begin{bmatrix} \cos(\alpha) & \cos(\alpha + \frac{2\pi}{3}) & \cos(\alpha + \frac{4\pi}{3}) \\ \sin(\alpha) & \sin(\alpha + \frac{2\pi}{3}) & \sin(\alpha + \frac{4\pi}{3}) \\ 1/2 & 1/2 & 1/2 \end{bmatrix}. \quad (37)$$

The MS-to-dqMS transformation matrix is given by

$$\mathbf{T}_{\text{MS-dqMS}} = \begin{bmatrix} \mathbf{T}_{\text{dq}}(\theta_e) & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{T}_{\text{dq}}(\theta_e) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{T}_{\text{dq}}(\theta_e) \end{bmatrix} \quad (38)$$

where $\mathbf{T}_{\text{dq}}(\theta_e)$ is given by

$$\mathbf{T}_{\text{dq}}(\theta_e) = \begin{bmatrix} \cos(\theta_e) & \sin(\theta_e) & 0 \\ -\sin(\theta_e) & \cos(\theta_e) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (39)$$

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